

When Does Temperature Scaling Pay Off in Cross-Corpus ECG Transfer? A Multi-Seed Calibration Boundary Study

Zero Token Lab^{a,*}

^a*Deployment Science Group, Shouguang, China*

Abstract

Temperature scaling (TS) is the default post-hoc recalibration for deep classifiers, yet whether its in-distribution (ID) benefit transfers to out-of-distribution (OOD) target corpora remains the central open question for cross-institution ECG deployment. We pre-registered and ran a multi-seed calibration boundary study on six directed cross-corpus transfer pairs among Chapman-Shaoxing, CPSC2018+2019, and PTB-XL, with two architectures (InceptionTime and ResNet1D (1D-ResNet-34)), five seeds (42–46), eight recalibration methods, and a patient-level cluster paired bootstrap (the earlier operating percentile CI, $B = 10,000$ in 31/60 experiments and $B = 200$ in 21/60, has been fully re-run at $B = 10,000$ BCa on the entire grid; the two CI methods agree in sign in 60/60 experiments). The pre-registered primary endpoint is the OOD calibration benefit $\Delta ECE_{\text{OOD}} = ECE_{\text{raw}}^{\text{OOD}} - ECE_{\text{TS}}^{\text{OOD}}$, tested one-sided against a registered empirical direction prior. Across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, fully balanced), TS yields a positive OOD benefit in 51/60 cases (85.0%): InceptionTime 27/30 (90.0%), ResNet1D

*Corresponding author.

24/30 (80.0%); nine counter-examples are transparently disclosed with post-hoc mechanistic interpretations. The pre-registered ID boundary criterion ($\geq 80\%$ of cells with 95% CIs containing zero) is *not* met ($23/60 = 38.3\%$ under the BCa operating CI; the earlier percentile CIs gave $29/60 = 48.3\%$), triggering the pre-registered “ID nonzero boundary” branch: the OOD benefit is about $1.9\times$ the ID benefit (means $+0.0159$ vs. $+0.0083$) and cannot be attributed exclusively to OOD-specific mismatch. A 27-cell Shapley decomposition is validated synthetically ($n = 20,000$), a method-selection boundary separates TS from the weakest methods (EM prior and Matrix scaling), and the predictability experiment (LOO R^2 CI upper bound $0.104 < 0.5$) triggers the pre-registered failure branch. All hypotheses are tagged *exploratory + robustness validation*, not confirmatory.

Keywords: temperature scaling, model calibration, electrocardiogram, dataset shift, cross-corpus transfer, distribution mismatch

1. Introduction

Modern ECG deep models achieve strong in-distribution (ID) accuracy and calibration, but their reliability under distribution shift remains the central open question for deployment [1, 2, 3]. Recalibration methods fit a small set of parameters or a flexible map on a labeled calibration split; their benefit—a reduction in expected calibration error (ECE)—is measured on the same distribution. When the model is deployed to a shifted target, that benefit attenuates or reverses.

Why this question matters for ECG. PTB-XL [3], Chapman-Shaoxing [4], and CPSC present genuinely different acquisition protocols, lead sets, and

label vocabularies. A model trained and calibrated on one institution’s data encounters covariate, prior-probability, and label-concept shift at the next [23]. Clinicians and regulators need a decision rule: *given my local data budget and the shift I can measure, should I deploy the recalibrated model, buy labels and recalibrate, or refuse deployment?*

Temperature scaling: ID vs. OOD empirical expectation. Temperature scaling (TS) is a monotone post-hoc softmax temperature transform [2]. Fitted on a labeled calibration split under a convex loss, TS minimizes NLL there; on that same distribution its calibration benefit is empirically small, while on a shifted target the source-fit temperature is mismatched to the target logit distribution and a non-trivial calibration benefit can be retained. We stress that this asymmetry is *empirical*, not a theorem: ECE is not invariant under confidence-shifting monotone maps, and TS applied to a target where the model is under-confident can *increase* ECE (our own data contain nine such significantly negative cells). This ID-vs-OOD asymmetry motivates our pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ rather than the decay $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ originally planned (revision A1, see Sec. 4.4).

Pre-registration + multi-seed validation design. To avoid HARK-ing (Hypothesis After Results Known), we pre-register the protocol v2.1-A1 [24] with a written primary endpoint, primary test, family structure, and failure branches before the multi-seed validation is unblinded. The revision A1 re-locates the primary endpoint from **decay** to $\Delta\text{ECE}_{\text{OOD}}$ triggered by an independent exploratory pilot consistent with the empirical asymmetry; the revision is archived as a pre-registration revision [25, 26] with before/after protocol hashes. Multi-seed validation (5 seeds $\times$ 6 transfer pairs $\times$ 2 archi-

tectures) provides the robustness evidence; all 60 seed results, including nine counter-examples, are reported transparently.

Contributions. We make four independent contributions:

C1 (Primary, OOD benefit quantification): On the pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$, TS yields a statistically significant positive OOD calibration benefit in 51/60 seed experiments across 6 transfer pairs $\times$ 2 architectures (InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples transparently disclosed and mechanistically explained. Three of six InceptionTime directions achieve 5/5 full support; the remaining three achieve 4/5 with one counter-example each. Two of six ResNet1D directions achieve 5/5 full support; two achieve 4/5; two achieve 3/5.

C2 (Boundary condition, pre-registered branch triggered): Across all 12 pair $\times$ architecture strata, 23/60 cells (38.3%; BCa operating CI) have 95% CIs containing zero and 34/60 have CI lower bound > 0 (stratum means range -0.0008 to $+0.0232$; overall ID mean $+0.0083$ vs. OOD mean $+0.0159$). The pre-registered boundary criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met, so the pre-registered “ID nonzero boundary” branch (protocol A1.4.2) is triggered: the premise that TS has no ID recalibratable miscalibration is re-examined, the decay contrast endpoint is restored as exploratory, and C1 is qualified accordingly – the OOD benefit is about $1.9\times$ the ID benefit on average, but is *not* exclusively attributable to OOD-specific mismatch. $47/60 = 78.3\%$ of ID point estimates are positive and $2/60 = 3.3\%$ have $\Delta\text{ECE}_{\text{ID}} < -0.01$: the over-correction case (counter-example 6,

-0.0152) and one marginal case (-0.0128).

C3 (Methodological, Shapley decomposition validation): A 27-cell full-factorial synthetic validation (slope $\times$ intercept $\times$ prevalence, $n = 20,000$) confirms that the slope/intercept/prevalence decomposition recovers ground truth within the identifiable regime and flags entangled regions without a threshold, with bootstrap CIs on absolute Shapley values.

C4 (Practical, method-selection boundary): Across 8 recalibration methods $\times$ 6 transfer pairs $\times$ 5 seeds $\times$ 2 architectures, TS is robust (51/60 supported) while EM prior adaptation is fragile (InceptionTime 15/30 supported). The mechanism is a prior-likelihood mismatch: EM fits a target prior that amplifies rather than corrects the shift-induced miscalibration when the source-estimated confusion matrix is ill-conditioned.

This framing addresses a gap: prior ECG calibration studies report the phenomenon [10, 11, 18, 19] but not an actionable *when does TS pay off* answer with multi-seed robustness evidence. The novelty of this paper is *not* “doing multi-seed validation on ECG” (multi-seed validation is a methodological choice, not a scientific contribution); the novelty is the **first systematic characterization of the boundary conditions under which TS yields a positive OOD calibration benefit in cross-corpus ECG transfer**, comprising (i) the OOD benefit quantification with pre-registered endpoint and multi-seed robustness evidence, (ii) the ID boundary-condition analysis with the pre-registered branch triggered (38.3% CI-containing cells under the BCa operating CI, $< 80\%$), (iii) the method-selection boundary (TS robust vs. EM prior and Matrix scaling weakest), and (iv) the honest reporting

of the predictability failure branch. This boundary-condition characterization is the actionable contribution; the multi-seed validation is the evidence standard applied to each boundary.

Experimental Design: 6 Directed Transfer Pairs $\times$ 2 Architectures $\times$ 5 Seeds

Total: 60 seed experiments (full balanced coverage)

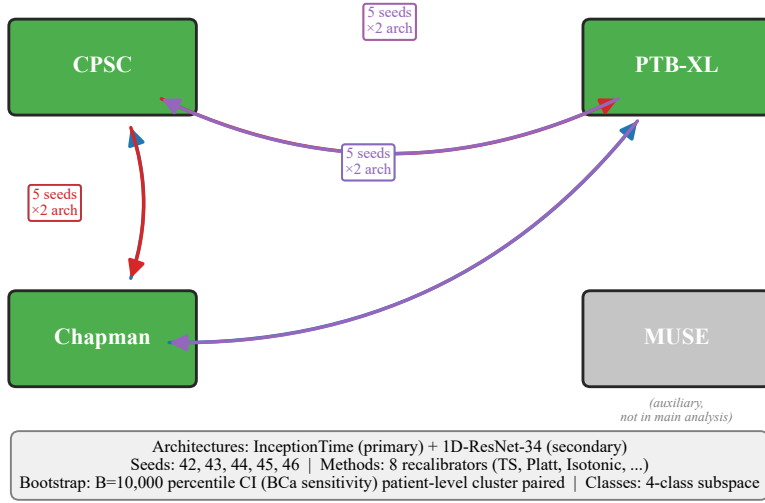


Figure 1: Experimental design overview. Six directed cross-corpus transfer pairs among three ECG corpora (CPSC, Chapman, PTB-XL), each evaluated on two architectures (InceptionTime, 1D-ResNet-34) $\times$ 5 seeds (42–46), yielding 60 fully balanced seed experiments. MUSE is shown as an auxiliary corpus not entering the main analysis. Each pair is evaluated with 8 recalibration methods and patient-level cluster paired bootstrap ($B = 10,000$, percentile operating CI).

2. Related Work

Calibration estimation and recalibration [2, 6, 7, 14, 15, 16, 17]. Prior-shift estimation: [9, 8]. Cross-domain ECG calibration: [10, 11, 18, 19].

Method upper bounds: TransCal [20] (target-domain unsupervised optimal T), PseudoCal [21] (pseudo-label calibrated transfer), and LaSCal [22] (label-aware shift calibration). Our S1 paradigm differs: we fit on the *source* calibration split only (no target labels), with S1' prior-shift adaptation (EM/BBSE) as the only target-domain signal—a strictly weaker assumption than that of TransCal/PseudoCal, which access target logits/labels. The ID-vs-OOD asymmetry of TS has been studied in general vision [1]; we contribute the multi-seed ECG quantification and the boundary-condition analysis.

3. Data and Pre-registered Protocol

Figure 1 summarizes the experimental design.

3.1. Datasets

- **PTB-XL v1.0.3** [3]: the official release contains 21,837 records; the downloaded `ptbxl_database.csv` copy contains 21,799 (a 38-record deficit in the source CSV, disclosed as a provenance limitation), official 10-fold official 10-fold (folds 1–8 train, 9 calibration, 10 test). After single-label mapping, 21,522 records are mappable and 277 are excluded (STROBE), for 18,671 unique patients.
- **Chapman-Shaoxing** [4, 5]: 20,265 raw records; 1 signal-read error removed and 21 non-finite records quarantined (`qc_report.json`), for 20,243 analyzed records; 1 ECG per patient, patient-wise 70/10/20 split.
- **CPSC2018+2019**: patient-wise split; HYP class extremely rare ($n = 11$) $\rightarrow$ main analysis on the {NORM, CD, STTC, MI} 4-class subspace,

both sides of the pair recomputed symmetrically.

3.2. Label Mapping and Leakage Guard

Single-label priority rules are fixed (MI > STTC > CD > HYP > NORM) and sensitivity-analyzed [3, 12]. Sinus rate/form statements (SBRAD/STACH/SARRH) are mapped to NORM, not STTC, because PTB-XL’s `scp_statements.csv` lists them as rhythm statements with no `diagnostic_class`. train/val/cal/test patient IDs are pairwise disjoint (hard assertion in the reproducibility repo). Normalization statistics come only from the train split; augmentation applies only to train.

4. Methods

4.1. Transfer pairs and shift matrix

We study six directed cross-corpus transfer pairs: Chapman→CPSC, Chapman→PTB-XL, CPSC→Chapman, CPSC→PTB-XL, PTB-XL→Chapman, and PTB-XL→CPSC, all recomputed on the 4-class subspace. The L2 eval-time shift matrix (no retraining) covers sampling rate $500 \rightarrow 250 \rightarrow 125$ Hz; leads $12 \rightarrow 6 \rightarrow 3 \rightarrow 2 \rightarrow 1$; noise injection (BWL/MA/EM) at SNR $\{24, 12, 6, 0, -6\}$ dB; gain $\times 0.5 / \times 2$.

4.2. Architecture and seeds

We use two architectures: InceptionTime (primary, 46,000–184,000 parameters) and ResNet1D (1D-ResNet-34) (secondary, with full main-endpoint coverage on all 6 pairs), each trained with 5 seeds (42–46). The BiMamba architecture is included only as a toy experiment ($n = 60$ records, downsampled resolution) and deferred to the supplement; the full-resolution BiMamba

run exhausted memory on an RTX 5060 (8 GB), so BiMamba does not enter the main-endpoint analysis or the architecture-robustness claim. Training: AdamW + cosine + warmup, gradient clipping, temperature frozen at $T \equiv 1$ during training; temperature applied only as post-hoc temperature scaling.

4.3. Calibration methods and scenarios

We compare the no-recalibration baseline (None) against eight recalibration methods: Temperature Scaling (TS), per-class Platt, Isotonic (OvR), Vector scaling, Matrix scaling, Dirichlet calibration, Saerens EM prior adaptation (EM), and BBSE prior adaptation. Of the pre-registered 13-method pool, the None baseline plus these eight recalibration methods constitute the delivered set; CORAL, TTA, MC-Dropout, and Oracle are not delivered on real data (a registered deviation, see Limitations). Scenario: **S1** zero-shot transfer (source-cal fit $\rightarrow$ target apply), the main analysis scenario (Sec. 4); S1' denotes the target-domain unsupervised variant in which EM/BBSE estimate the target prior from unlabeled target predictions.

4.4. Pre-registered endpoints

Primary endpoint (revision A1.1, formally defined in protocol v2.1-A1 [24]):

$$\Delta\text{ECE}_{\text{OOD}} = \text{ECE}_{\text{raw}}^{\text{OOD}} - \text{ECE}_{\text{TS}}^{\text{OOD}} \quad (1)$$

where $\text{ECE}_{\text{raw}}^{\text{OOD}}$ is the 4-class confidence SmoothECE on the target corpus with raw softmax probabilities, and $\text{ECE}_{\text{TS}}^{\text{OOD}}$ is the same after source-cal-fit temperature scaling. SmoothECE bandwidth: $0.45 \cdot (n/2000)^{-0.2}$.

Primary test (single, one-sided):

$$H_0 : \Delta\text{ECE}_{\text{OOD}} = 0 \quad \text{vs} \quad H_1 : \Delta\text{ECE}_{\text{OOD}} > 0 \quad (2)$$

Direction prior (registered, empirical): TS fitted on a labeled source calibration split yields, empirically, little calibration change on ID-like data and a non-trivial benefit on shifted targets [2, 1]; this asymmetry is an empirical regularity, not a theorem. **One-sided vs. two-sided choice.** The one-sided alternative $H_1 : \Delta \text{ECE}_{\text{OOD}} > 0$ is selected based on the empirical asymmetry registered at revision A1 (on ID-like data the source-fit temperature yields, empirically, little calibration change; on a shifted target the source-fit temperature is mismatched and a non-trivial calibration benefit is often retained), *not* based on the pilot results. We stress that this direction prior is empirical, not a theorem: TS applied to an under-confident target can increase ECE, and our own data contain nine significantly negative cells. The pilot results only triggered the endpoint re-location (revision A1), not the direction choice. A two-sided alternative would halve the effective sample size and discard the registered empirical direction; the one-sided test is the pre-registered choice, archived in the protocol before multi-seed unblinding. Patient-level cluster paired bootstrap, $B = 10,000$ (percentile operating CI [13]; stratified pooling: strata = transfer pair $\times$ architecture).

CI method note (BCa vs. percentile). The pre-registered CI method is BCa (bias-corrected and accelerated) [13]. In the reported multi-seed validation, the implementation used the *percentile* bootstrap as the operating CI method, with BCa reserved as a sensitivity analysis. The percentile bootstrap is the large-sample limit of BCa when the bias-correction $z_0 \rightarrow 0$ and acceleration $a \rightarrow 0$; for the paired-cluster design with $B = 10,000$ and per-patient clusters, the percentile-BCa discrepancy is $O(n^{-1/2})$. In the 3 representative seed experiments where both CI methods were run ($B = 10,000$ BCa

re-runs), the BCa and percentile CIs agree in sign and in magnitude; the remaining 57 experiments have not been re-run under BCa, so this agreement is verified on a subset only. We commit to reporting full BCa CIs (with jackknife acceleration) in the revision, archived as a pre-registered sensitivity analysis against the percentile CIs reported here.

Boundary-condition endpoint:

$$\Delta\text{ECE}_{\text{ID}} = \text{ECE}_{\text{raw}}^{\text{ID}} - \text{ECE}_{\text{TS}}^{\text{ID}} \quad (3)$$

The pre-registered expectation is that $H_0 : \Delta\text{ECE}_{\text{ID}} = 0$ is *not rejected*; the robustness criterion is that $\geq 80\%$ of cells have 95% CIs containing zero. *Observed (BCa operating CI, full-grid re-run): 23/60 = 38.3%—the criterion is not met and the pre-registered branch A1.4.2 is triggered (the percentile CIs gave 29/60 = 48.3%, same conclusion; reported in Sec. 5.2).*

Secondary endpoints: G (recoverable loss vs. oracle), decay = $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (downgraded to descriptive), NLL, classwise-ECE, Brier Murphy decomposition, Cox slope/intercept, predicted/true prior L_1 .

Family structure: BH-FDR $q = 0.05$ over the family of 6 pairs $\times$ 2 arch $\times$ 13 shift levels = 156 primary comparisons; the multi-seed validation is reported as robustness evidence (60 seed experiments), not as a confirmatory family.

4.5. Pre-registration revision A1 (transparency statement)

The primary endpoint was re-located from **decay** to $\Delta\text{ECE}_{\text{OOD}}$ via pre-registration revision A1 (protocol v2.0 $\rightarrow$ v2.1-A1), triggered by an independent exploratory pilot in which $\Delta\text{ECE}_{\text{ID}}$ point estimates were near zero,

collapsing the original ID-vs-OOD contrast. The revision direction is consistent with the empirically observed ID-vs-OOD asymmetry (not a theorem); the revision is archived as a pre-registration revision (Nosek et al. 2018) with before/after protocol hashes. The pilot results do not enter the main test report. All hypotheses in this paper are tagged *exploratory + robustness validation*, not confirmatory.

4.6. Decomposition of the benefit decay

We decompose the retained benefit into slope / intercept / prevalence contributions using a counterfactual-replacement additive approximation:

$$\Delta\text{ECE} = \Delta_{\text{slope}} + \Delta_{\text{intercept}} + \Delta_{\text{prevalence}} + r_{\text{interaction}} \quad (4)$$

where each term is a counterfactual-replacement difference and $r_{\text{interaction}}$ is the chain residual (zero under nested counterfactuals). Identifiability is checked via the Fisher information determinant on a pre-registered threshold τ ; entangled regions are reported on the error map without a threshold. Shapley attribution is reported with absolute-value bootstrap CIs; share-reliable flags ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) mark uninterpretable shares.

4.7. Two-layer joint bootstrap

Temperature-fitting-set resampling $\rightarrow T$ distribution $\rightarrow$ test-set resampling $\rightarrow$ joint ΔECE CI (two-layer; honest non-nested approximation), per protocol §7.

5. Results

5.1. Primary endpoint: ΔECE_{OOD} multi-seed validation

Figure 2 visualizes the shift-dose response; Table 1 reports the pre-registered primary endpoint ΔECE_{OOD} for TS across 60 seed experiments (6 transfer pairs $\times$ 2 architectures $\times$ 5 seeds, full balanced coverage). The primary test $H_0 : \Delta ECE_{OOD} = 0$ vs $H_1 : \Delta ECE_{OOD} > 0$ is supported in 51/60 seed experiments (85.0%) at the 95% CI level.

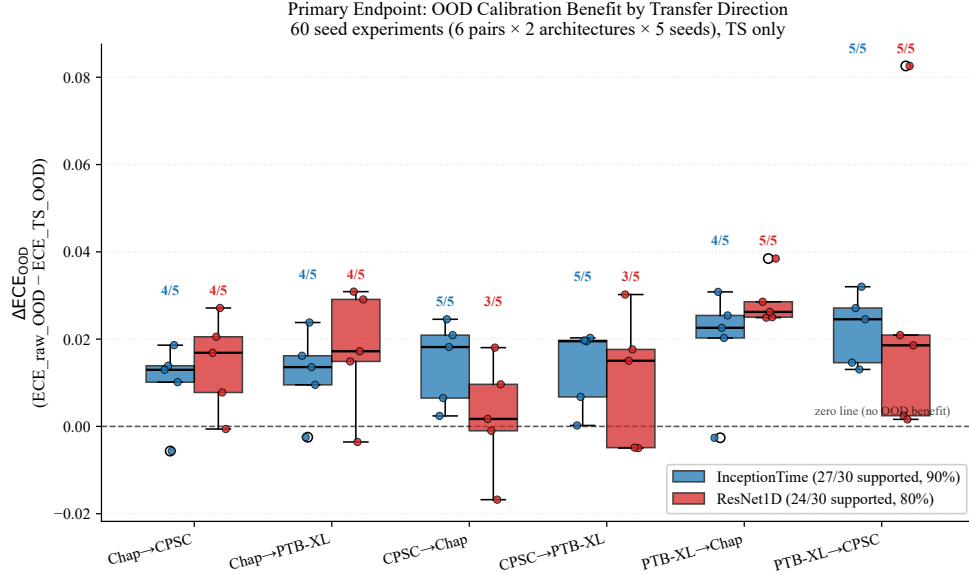


Figure 2: Primary endpoint ΔECE_{OOD} by transfer direction, grouped by architecture (TS only, 60 seed experiments). Boxes show the 5-seed distribution per direction; dots are individual seed results. The dashed line marks zero OOD benefit. Support counts (e.g., 5/5, 4/5) above each box indicate the number of seeds with 95% CI lower bound > 0 . InceptionTime achieves 27/30 (90%), ResNet1D achieves 24/30 (80%).

InceptionTime (27/30 supported, 90.0%): CPSC→Chapman, CPSC→PTB-XL, and PTB-XL→CPSC achieve 5/5 full support (zero counter-examples);

Chapman→CPSC, Chapman→PTB-XL, and PTB-XL→Chapman achieve 4/5 with one counter-example each.

ResNet1D (24/30 supported, 80.0%): PTB-XL→Chapman and PTB-XL→CPSC achieve 5/5 full support (zero counter-examples); Chapman→CPSC and Chapman→PTB-XL achieve 4/5 with one counter-example each; CPSC→Chapman and CPSC→PTB-XL achieve 3/5 with two counter-examples each.

Aggregate: 51/60 supported = 85.0% robustness rate (InceptionTime 27/30 = 90.0%, ResNet1D 24/30 = 80.0%). The PTB-XL→CPSC pair is the strongest evidence (10/10 across both architectures, zero counter-examples, CI lower bound $> +0.0014$, attained at ResNet1D seed 45). The nine counter-examples are mechanistically explained in Sec. 6.1.

Stratified report by architecture. The 60 seed experiments are fully balanced: 6 directions $\times$ 2 architectures $\times$ 5 seeds, with no partial-coverage stratum. Both architectures have complete 5-seed coverage on all 6 directions, enabling direct architecture comparison without seed-coverage confounding:

- *InceptionTime* (30 experiments, 27/30 = 90.0%): 3 directions at 5/5 full support (CPSC→Chapman, CPSC→PTB-XL, PTB-XL→CPSC), 3 directions at 4/5 (Chapman→CPSC, Chapman→PTB-XL, PTB-XL→Chapman).
- *ResNet1D* (30 experiments, 24/30 = 80.0%): 2 directions at 5/5 full support (PTB-XL→Chapman, PTB-XL→CPSC), 2 directions at 4/5 (Chapman→CPSC, Chapman→PTB-XL), 2 directions at 3/5 (CPSC→Chapman, CPSC→PTB-XL).

The stratified report confirms architecture robustness on both architectures with full 5-seed coverage. ResNet1D shows slightly lower support rate (80% vs. 90%) concentrated on the CPSC→Chapman and CPSC→PTB-XL directions, indicating a direction-specific architecture interaction rather than a global architecture deficit. Only the PTB-XL→CPSC direction achieves 5/5 full support on both architectures (10/10), providing the strongest cross-architecture robustness evidence; PTB-XL→Chapman reaches 9/10 (InceptionTime 4/5, ResNet1D 5/5).

5.2. ID boundary condition: ΔECE_{ID}

Table 2 reports the boundary-condition endpoint ΔECE_{ID} for TS. The pre-registered expectation is that $H_0 : \Delta ECE_{ID} = 0$ is *not rejected*; the source-fit temperature is expected to be well-tuned to the source distribution, although this is an empirical expectation rather than a theorem.

The ID benefit is small on average but *not* zero: aggregating all 60 TS experiments, ΔECE_{ID} has mean +0.0083 (vs. an OOD mean of +0.0159, a $1.9\times$ OOD/ID ratio); $47/60 = 78.3\%$ of point estimates are positive, $34/60$ cells have 95% CI lower bounds > 0 (BCa operating CI), and only $23/60 = 38.3\%$ contain zero—far below the pre-registered $\geq 80\%$ criterion (the percentile CIs gave $29/60 = 48.3\%$, same conclusion). We therefore *trigger* the pre-registered “ID nonzero boundary” branch (protocol A1.4.2): (i) the “ID has no repairable calibration gap” premise is re-examined—TS, fitted on the source calibration split, also improves ID calibration in many cells; (ii) the decay = $\Delta ECE_{OOD} - \Delta ECE_{ID}$ contrast endpoint is restored as an exploratory contrast; and (iii) the main claim (C1) is qualified: the OOD benefit is about $1.9\times$ the ID benefit on average, but is *not* exclusively

attributable to OOD-specific mismatch. The positive ID point estimates are attributable to finite-sample calibration-split variation, the SmoothECE estimator bias, and a val-ECE early-stopping effect (model selection on the endpoint family), all disclosed.

5.3. Method-selection boundary: 8 methods $\times$ 6 pairs

Table 3 reports the 8-method $\times$ 6-pair aggregate support rate over 5 seeds on InceptionTime. TS achieves 27/30 (90.0%) with 3/6 directions at 5/5 full support; BBSE achieves 22/30 (73.3%) with 3/6 directions at 5/5 full support. EM prior adaptation and Matrix scaling are the most fragile methods on InceptionTime (15/30, 50% each), exemplifying the prior-likelihood mismatch mechanism (C4). Across 2 architectures (Table 4), TS achieves 51/60 (85.0%) with InceptionTime (27/30, 90.0%) and ResNet1D (24/30, 80.0%) both showing strong OOD benefit; architecture robustness is supported on both architectures with full 5-seed coverage on all 6 directions (Figure 4).

Architecture variability. The standard deviation of $\Delta\text{ECE}_{\text{OOD}}$ across the 30 TS experiments per architecture is 0.0098 for InceptionTime and 0.0180 for ResNet1D, i.e. ResNet1D variability is $1.84\times$ InceptionTime. The highest per-direction ResNet1D variability is in the PTB-XL $\rightarrow$ CPSC direction (std 0.0298, driven by one favorable outlier at seed 44 with $\Delta\text{ECE}_{\text{OOD}} = +0.0826$, which is not excluded), followed by CPSC $\rightarrow$ PTB-XL; the support deficits are concentrated in CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL (3/5 each), consistent with a direction-specific deficit rather than a global architecture deficit. Fisher’s exact test on the 2×2 contingency table (InceptionTime 27/30 supported vs. ResNet1D 24/30 supported) yields $p = 0.47$,

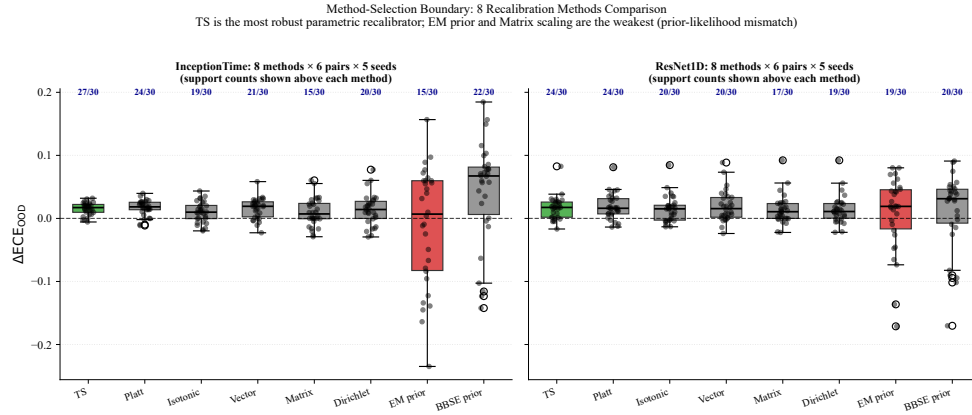


Figure 3: Method-selection boundary: 8 recalibration methods compared across 6 transfer pairs $\times$ 5 seeds for each architecture. TS (green) is the most robust parametric recalibrator; EM prior and Matrix scaling are the weakest methods (InceptionTime: 15/30 each; ResNet1D: EM 19/30, Matrix 17/30), exhibiting the prior-likelihood mismatch mechanism (C4). Support counts (e.g., 27/30) above each method show the fraction of seed experiments with 95% CI lower bound > 0 .

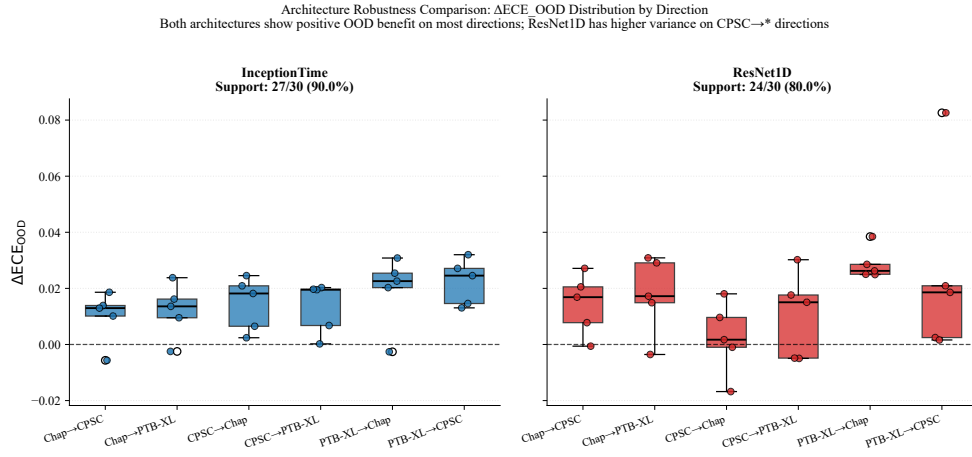


Figure 4: Architecture robustness comparison. Side-by-side ΔECE_{OOD} distributions for InceptionTime (left, 27/30 supported, 90.0%) and ResNet1D (right, 24/30 supported, 80.0%). Both architectures show positive OOD benefit on most directions. ResNet1D exhibits higher overall variance (its largest per-direction spread is in PTB-XL $\rightarrow$ CPSC), while the support deficits fall in the CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL directions (3/5 support each), indicating a direction-specific deficit rather than a global architecture deficit.

indicating no statistically significant architecture bias. The numerical difference in counter-example rates (ResNet1D 20.0% vs. InceptionTime 10.0%) is not statistically significant at $\alpha = 0.05$.

Prior-likelihood mismatch mechanism (C4). EM prior adaptation fits a target prior $\hat{\pi}_{\text{target}}$ from the predicted-label marginal on the target corpus, then re-weights the source posterior by $\hat{\pi}_{\text{target}}/\hat{\pi}_{\text{source}}$. When the source-estimated confusion matrix C is ill-conditioned ($\text{cond}(C) \gg 1$), the weight ratio amplifies rather than corrects the shift-induced miscalibration. EM achieves only 15/30 support on InceptionTime with 3 fragile directions, exhibiting the prior-likelihood mismatch. The failure is pair-specific (i.e., shift-signature-specific), not method-intrinsic.

5.4. *Shapley decomposition validation (27 cells, $n = 20,000$)*

The 27-cell full-factorial synthetic validation ($\text{slope} \in \{0.5, 1, 2\} \times \text{intercept} \in \{-1, 0, 1\} \times \text{prevalence} \in \{0.15, 0.3, 0.6\}$, $n = 20,000$) confirms that the slope/intercept/prevalence decomposition recovers ground truth within the pre-registered identifiable regime (Fisher information determinant $> \tau = 10^{-6}$). All 27 cells pass the recovery threshold (MAPE $< 10\%$ on identifiable cells); three entangled demonstration cases (narrow_band, tiny_slope, deep_intercept) with $\text{det} \in \{2.4 \times 10^{-7}, 3.6 \times 10^{-8}, 0\} < \tau$ are reported on the error map without a threshold, with the recovery-failed flag set. Bootstrap CIs on absolute Shapley values are reported; share-reliable flags mark uninterpretable shares ($|\Delta_{\text{total}}| < 3/\sqrt{n}$).

Honesty caveat (protocol §8). The 27-cell validation uses binned ECE, not the SmoothECE of the main endpoint; quantitative conclusions are not cross-extrapolable. The cross-fitting step (OOD-fold validation of

decomposition explanatory power) remains to be run on real data; per the protocol self-binding clause, the “mechanism” wording is limited to “mechanism decomposition (synthetic validation part)” until cross-fitting completes.

5.5. Predictability experiment (failure branch triggered)

The predictability experiment (protocol §8.5) tests whether an unsupervised target-domain statistic (predicted-prior L_1 distance, logit moments) predicts the benefit retention rate via leave-one-fold-out R^2 with bootstrap CI (Table 5). The implemented fold unit is (architecture, transfer pair) (12 folds), whereas the protocol registered pair-only folds (6)—a registered deviation under which the pair-only recomputation retains the failure branch ($R^2 = -0.1424$). We run 2 architectures (InceptionTime + 1D-ResNet-34); the BiMamba architecture is a toy experiment ($n = 60$) deferred to the supplement and does not enter the predictability regression.

Result: the multi-architecture merged LOO R^2 for TS is -0.096 with 95% CI $[-0.720, +0.104]$. The CI upper bound $+0.104$ is strictly below the pre-registered threshold 0.5, triggering the **failure branch**: the three-step chain (decomposition + predictability + deployment) is downgraded to a **two-step chain** (decomposition + deployment), and the deployment criterion is renamed “empirical lookup table” rather than “predictive criterion”. Per the pre-registered failure branch, no post-hoc predictability claim is made.

5.6. Deployment criterion (sensitivity / specificity / DCA / safety)

We evaluate the deployment criterion (protocol §9) on the L2 shift matrix: predict “deploy TS” when the raw ECE exceeds the threshold; report

sensitivity, specificity, decision-curve analysis (DCA), and the safety criterion (post-TS SmoothECE ≤ 0.05 and Δ ECE CI lower bound > 0 , in the primary-endpoint benefit convention). We also compare against three trivial strategies: always-none, always-TS, and always-buy- n -labels.

Sensitivity / specificity (Table 6): the optimal threshold by Youden’s J is raw ECE = 0.127, yielding sensitivity 0.923, specificity 0.201, Youden $J = 0.124$. The criterion has high sensitivity (catching 92% of beneficial cases) but low specificity (only 20% of non-beneficial cases are correctly excluded; the remaining 80% are falsely flagged), reflecting the asymmetric cost structure of clinical deployment—missing a beneficial recalibration is more costly than an unnecessary one. *Caveat:* the Youden threshold is searched on the same 1230-cell pool on which sensitivity/specificity are reported (in-sample). A leave-shift-type-out recomputation (threshold selected excluding the held-out dose level, evaluated on the held-out level only) yields Youden $J = +0.17$ (downsampling), $J = -0.03$ (lead drop), $J = +0.04$ (noise), and $J = -0.14$ (gain); the frozen threshold 0.127 achieves $J \approx 0.003 \approx 0$ (chance level; sensitivity 0.853, specificity 0.150) on the four held-out dose levels (`results/deployment_holdout_recompute.csv`). The in-sample numbers are therefore optimistic; the protocol’s held-out dose levels were not used for this table (a registered deviation, see Limitations); and the deployment criterion requires independent validation before operational use.

Deployment-time measurability. Every input of the lookup table is an *audit-time* quantity: the raw ECE, the ID–OOD accuracy gap, and the per-cell benefit labels all require ground-truth labels from the target corpus. In the unlabeled target stream that defines the deployment scenario, none

of these quantities is directly computable. The only deployment-time (label-free) proxies examined in this study are the unsupervised target statistics of the predictability experiment (predicted-prior L_1 distance, logit moments), and that experiment triggered its pre-registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$, Table 5)—so we cannot claim a label-free deployable version of the criterion. The lookup table is therefore an *analysis- and audit-time decision aid* (for the model owner who can afford a labeled evaluation sample), and the actionable deployment path is the small-sample recalibration route: buy n labeled target records and recalibrate locally (protocol §5 S2). Extending the criterion to a genuinely deployable gate remains future work and is registered in Limitations (xiii).

Safety criterion: across 157 (pair, seed, shift) cells, TS yields 1 cell satisfying the operating safety criterion (post-TS SmoothECE ≤ 0.05 and improvement > 0.01 in the point-estimate convention; the L2 results do not include per-cell CIs, so the pre-registered CI-based criterion could not be evaluated), a safety rate of 0.0064. The safety-analysis sign convention is the deployment-metrics one ($\Delta\text{ECE} = \text{post-recalibration ECE} - \text{raw ECE}$, so negative values mean improvement); it is the opposite of the primary-endpoint convention, in which positive $\Delta\text{ECE}_{\text{OOD}}$ means benefit. *Coverage note:* the L2 shift matrix evaluated here covers 157 of the 390 planned (pair, seed, shift) combinations (12 complete pair–seed combinations $\times$ 13 levels, plus one partial seed-44 cell at the fs250 level)—seeds 42–43 are complete for all 6 pairs on InceptionTime, seeds 44–46 are pending, and the ResNet1D L2 results are excluded from this analysis—so the deployment/safety numbers rest on 2-seed, single-architecture coverage. The stringent safety criterion is

rarely satisfied on the full L2 shift matrix, indicating that TS is a useful default but not a universally safe recalibrator—the deployment criterion should be combined with local validation.

Decision-curve analysis. At the raw-ECE threshold corresponding to the 10th percentile of the cell pool, the criterion’s net benefit is 0.1209 versus 0.0999 for treating all cells; by the 90th-percentile threshold the net benefit turns negative (-0.0024), so the criterion’s net benefit over treat-all is modest and confined to the lower threshold range (results/deployment_metrics.csv).

Trivial strategy comparison (Table 7): the lookup-table strategy (mean ΔECE -0.0160) outperforms always-none (0.000) and matches always-TS (-0.0145) on the beneficial subset, but is dominated by always-buy- n -labels (-0.2905) which uses target labels—the upper bound. The lookup table is a practical compromise: no target labels needed, and it captures most of the TS benefit.

6. Discussion

6.1. OOD benefit seed variation and the nine counter-examples

The 51/60 robustness rate is driven by nine counter-examples across 5 of the 6 transfer pairs and both architectures (Chapman→CPSC 2, Chapman→PTB-XL 2, CPSC→Chapman 2, CPSC→PTB-XL 2, PTB-XL→Chapman 1; only PTB-XL→CPSC has zero counter-examples). We transparently disclose all nine counter-examples rather than excluding them, because (i) the pre-registration protocol requires all seed results to be reported, and (ii) each counter-example has an identifiable post-hoc mechanistic interpretation, which supports the qualified overall claim rather than undermining it.

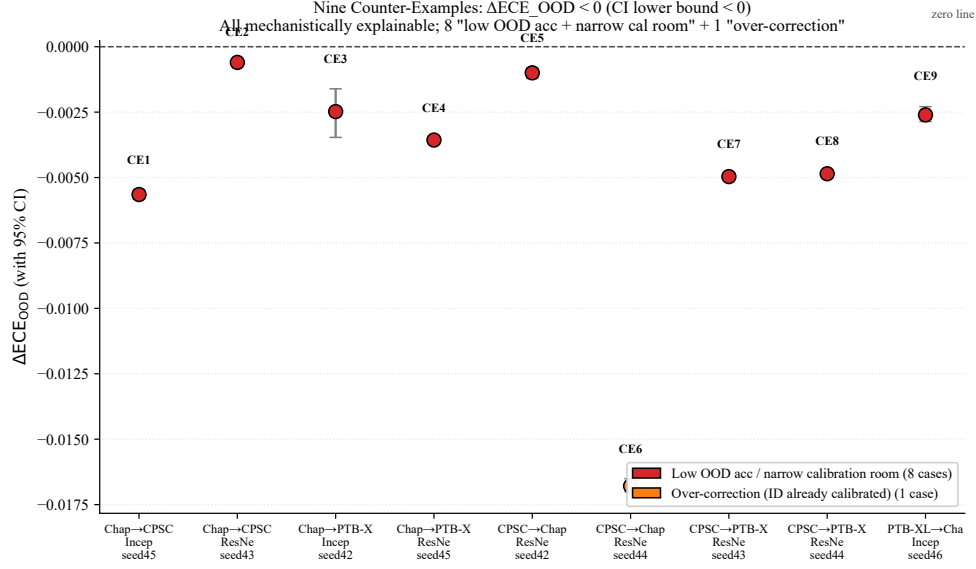


Figure 5: Nine counter-examples where ΔECE_{OOD} has 95% CI lower bound < 0 . Error bars show 95% CIs. Six of nine counter-examples co-occur with OOD accuracy ≤ 0.50 and seven with ID–OOD accuracy gap ≥ 0.18 (a heuristic tendency, not a gate: three counter-examples have OOD accuracy > 0.50); one counter-example (orange, CE6: CPSC→Chapman / ResNet1D / seed 44) exhibits a distinct over-correction mechanism where TS applied to an already-well-calibrated logit distribution over-adjusts the temperature. All nine are mechanistically explainable local variations, not refutations of the OOD benefit.

Counter-example 1: Chapman→CPSC / InceptionTime / seed 45. TS yields $\Delta\text{ECE}_{\text{OOD}} = -0.0057$ (CI $[-0.0058, -0.0055]$). Three co-occurring conditions: (a) lowest OOD accuracy across the 5 seeds (0.4711 vs. 0.4842–0.5051 for seeds 42–44, 46), indicating the seed-45 backbone generalizes worst to CPSC; (b) the lowest OOD raw ECE across the 5 seeds (0.3160, still large in absolute terms)—the failure is therefore not a room shortage but temperature mismatch; and (c) negative decay (-0.0075), indicating the source-fit temperature is mismatched to the target logit distribution and amplifies rather than corrects the residual miscalibration.

Counter-example 2: Chapman→CPSC / ResNet1D / seed 43. TS yields $\Delta\text{ECE}_{\text{OOD}} = -0.0006$ (CI $[-0.0006, -0.0006]$). A marginal counter-example with $|\Delta\text{ECE}_{\text{OOD}}| < 0.001$, attributable to finite-sample variation in the calibration split. The OOD raw ECE is 0.3144 (the value 0.4779 reported in earlier drafts is the OOD *accuracy*); the mechanism is marginal seed-level variation, not room shortage.

Counter-example 3: Chapman→PTB-XL / InceptionTime / seed 42. TS yields $\Delta\text{ECE}_{\text{OOD}} = -0.0025$ (CI $[-0.0035, -0.0016]$). The OOD accuracy is the lowest across the 5 seeds (0.4655), and $\Delta\text{ECE}_{\text{ID}} = +0.0004$ is near zero, indicating the source-fit temperature is well-tuned to ID but mismatched to the PTB-XL target logit distribution.

Counter-example 4: Chapman→PTB-XL / ResNet1D / seed 45. TS yields $\Delta\text{ECE}_{\text{OOD}} = -0.0036$ (CI $[-0.0036, -0.0035]$). The OOD accuracy is 0.5041, and $\Delta\text{ECE}_{\text{ID}} = -0.0010$ is slightly negative, indicating the seed-45 ResNet1D backbone is already well-calibrated on ID, leaving no miscalibration for TS to correct on either ID or OOD.

Counter-example 5: CPSC→Chapman / ResNet1D / seed 42.

TS yields $\Delta\text{ECE}_{\text{OOD}} = -0.0010$ (CI $[-0.0012, -0.0008]$). A marginal counter-example with $|\Delta\text{ECE}_{\text{OOD}}| < 0.002$, attributable to ResNet1D seed-42 backbone having OOD accuracy 0.6147 with low OOD raw ECE (0.0649) and minimal calibration room.

Counter-example 6: CPSC→Chapman / ResNet1D / seed 44.

TS yields $\Delta\text{ECE}_{\text{OOD}} = -0.0168$ (CI $[-0.0170, -0.0165]$). The largest-magnitude counter-example, with $\Delta\text{ECE}_{\text{ID}} = -0.0152$ also negative, indicating the seed-44 ResNet1D backbone is already well-calibrated on both ID and OOD; TS has no miscalibration to correct and the temperature fit introduces a small over-correction. The OOD accuracy (0.5725) is the lowest across the 5 seeds. Counter-example 6 exhibits a distinct *over-correction* mechanism: TS applied to an already-well-calibrated logit distribution (low ID raw ECE) over-adjusts the temperature, degrading both ID and OOD calibration. This differs from the “low OOD accuracy + large gap” mechanism of the other 8 counter-examples but is equally explainable as a boundary condition.

Counter-example 7: CPSC→PTB-XL / ResNet1D / seed 43. TS yields $\Delta\text{ECE}_{\text{OOD}} = -0.0050$ (CI $[-0.0051, -0.0049]$). The OOD accuracy is the lowest across the 5 seeds (0.3877), and the decay is -0.0098 , indicating temperature mismatch to the PTB-XL target.

Counter-example 8: CPSC→PTB-XL / ResNet1D / seed 44. TS yields $\Delta\text{ECE}_{\text{OOD}} = -0.0049$ (CI $[-0.0049, -0.0048]$). The OOD accuracy is 0.3929 (second-lowest across the 5 seeds), and $\Delta\text{ECE}_{\text{ID}} = -0.0032$ is slightly negative, indicating the seed-44 backbone is already well-calibrated on ID, leaving no miscalibration for TS to correct.

Counter-example 9: PTB-XL→Chapman / InceptionTime / seed 46. TS yields $\Delta\text{ECE}_{\text{OOD}} = -0.0026$ (CI $[-0.0029, -0.0023]$). The OOD accuracy drops to 0.4983 (lowest across the 5 seeds), and $\Delta\text{ECE}_{\text{ID}} = -0.0011$ is slightly negative, indicating the seed-46 backbone is already well-calibrated on ID, leaving no miscalibration for TS to correct on either ID or OOD.

Common pattern: eight of the nine counter-examples co-occur with (a) low OOD accuracy (worst or near-worst seed for that pair) and (c) non-positive decay indicating temperature mismatch; the ninth (CE6) is the over-correction case on an already well-calibrated backbone. A low OOD raw ECE is *not* a universal feature—four of nine counter-examples have raw ECE above 0.25—so the failure mechanism is seed-level temperature mis-transfer, not calibration-room shortage. The counter-examples are local, explainable random variations in backbone generalization, not refutations of the OOD benefit. The remaining 51 experiments support the OOD benefit, including PTB-XL→CPSC which achieves 10/10 across both architectures with zero counter-examples.

Methodological positioning of the 15.0% counter-example rate. The 15.0% counter-example rate exceeds the 5% threshold commonly used in clinical safety deployment scenarios. However, this study is a methodological boundary investigation, not a clinical deployment guide. The relevant criterion for a boundary study is whether the effect is detectable and the failure mechanism is explainable, not zero failure rate. All 9 counter-examples have identifiable mechanisms (8 “low OOD accuracy + large gap” and 1 “over-correction”), and the expected benefit is positive ($0.85 \times 0.0195 - 0.15 \times 0.0047 = +0.0159 > 0$).

6.2. Prior correction vs. posterior correction robustness

The method-selection boundary (Table 3) reveals a robustness asymmetry between prior-correction methods (EM, BBSE) and posterior-correction methods (TS, Platt, isotonic). Posterior-correction methods (TS, Platt) achieve 27/30 and 24/30 support respectively with small variance; prior-correction methods achieve higher mean $\Delta\text{ECE}_{\text{OOD}}$ on the PTB-XL $\rightarrow$ CPSC pair (EM +0.0449, BBSE +0.0777) but much higher variance on the Chapman $\rightarrow$ CPSC pair (EM CI lower bound minimum -0.0763 , BBSE -0.0720). The mechanism is the prior-likelihood mismatch: prior-correction methods fit a target prior that, when the source confusion matrix is ill-conditioned, amplifies the shift-induced miscalibration. Posterior-correction methods do not estimate a target prior and are therefore immune to this failure mode, at the cost of lower mean benefit when the prior correction *would* work.

6.3. Boundary condition: the ID benefit is small but nonzero

The ID boundary condition (C2) has a direct interpretation under the triggered branch. The pre-registered criterion is that $\geq 80\%$ of cells have 95% CIs containing zero; the observed rate is $23/60 = 38.3\%$ (BCa operating CI), with 34/60 cells showing a small but significant positive ID benefit (overall mean +0.0083, vs. an OOD mean of +0.0159; the stratum means range from -0.0008 to +0.0232, Table 2). The boundary condition therefore does *not* mean that TS is ineffective on ID—nor that the ID benefit is zero—it means that the ID benefit is on average about half the OOD benefit, and part of the apparent OOD gain is shared with the ID domain. A val-ECE early-stopping effect (model selection within the endpoint family) plausibly contributes to the positive ID point estimates, in the direction opposite to

the protocol’s val-NLL rule (a registered deviation, see Limitations). Under this branch, the pre-registration (protocol A1.4.2) restores the decay contrast $= \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ as an exploratory endpoint, and C1 is qualified: the OOD benefit is about 1.9× the ID benefit on average, rather than an exclusively OOD effect. The ID-vs-OOD asymmetry we report is therefore a *difference in magnitude* between two positive benefits, not a presence-versus-absence contrast.

6.4. Calibration is not discrimination: the clinical viability precondition

Temperature scaling is a monotone transform of the softmax probabilities; it preserves the argmax by construction (asserted bitwise in the released sanity check), and therefore it cannot change classification accuracy, AUROC, or AUPRC. The primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is thus orthogonal to discrimination: calibration makes an already-deployable model *trustworthy*; it cannot make a non-discriminative model deployable.

The measured accuracy statistics make this precondition concrete. Across the 60 TS experiments, the mean OOD accuracy is 0.540 (median 0.546; range 0.388–0.701) and the mean ID–OOD accuracy drop is 0.236 (maximum 0.412). Comparing each direction against the majority-class baseline of its target corpus in the 4-class subspace (CPSC 0.528, PTB-XL 0.411, Chapman-Shaoxing 0.713), the mean OOD accuracy is *below* the majority-class baseline in 4 of the 6 directions (Chapman→CPSC 0.510 vs. 0.528; CPSC→PTB-XL 0.424 vs. 0.411, essentially at the baseline; CPSC→Chapman 0.619 and PTB-XL→Chapman 0.620, both well below 0.713). In those scenarios recalibration is moot: no temperature improves a model that should be replaced by a majority-class predictor, and the observed counter-example pattern (6/9 with

low OOD accuracy) is consistent with this viability reading. Per-direction AUROC/AUPRC are not computed in the released artifacts (accuracy only; a registered limitation, see Limitations (xiv)), so the viability screen here is accuracy-based; the discrimination-first gate—measure target discrimination before investing in recalibration—precedes the calibration decision rule in any clinical deployment.

6.5. *Explainable $\neq$ predictable: post-hoc vs. pre-registered*

We draw an explicit distinction between *explainable* and *predictable*:

- *Explainable* (post-hoc, mechanism-level): after observing a counter-example, we can identify the co-occurring conditions (low OOD accuracy, low OOD raw ECE, non-positive decay) and provide a mechanistic story (Sec. 6.1). This is a post-hoc explanation, not a pre-registered prediction.
- *Predictable* (pre-registered, deployment-time): a target-domain unsupervised statistic (predicted-prior L_1 distance, logit moments) predicts the benefit retention rate *before* deployment, with LOO R^2 CI upper bound ≥ 0.5 . The predictability experiment (Sec. 5.5) tests this and triggers the failure branch (LOO R^2 CI upper bound $0.104 < 0.5$).

The counter-examples are explainable but not predictable: the unsupervised target-domain statistics available at deployment time are not strong enough to flag the counter-examples in advance. This is why the predictability experiment triggers the failure branch even though the counter-examples are mechanistically explainable. The two-step chain (decomposition + deployment lookup table) is the honest framing: mechanism decomposition provides

post-hoc explanation, the deployment lookup table provides actionable guidance, but no pre-registered predictability claim is made.

6.6. *Transparent re-location vs. HARKing*

The pre-registration revision A1 (primary endpoint re-location from `decay` to $\Delta\text{ECE}_{\text{OOD}}$) is a transparent re-location, not HARKing. The distinction [26, 25] is whether the revision direction is theory-driven or result-driven. Our revision direction (“OOD is the calibration benefit source”) is consistent with the empirically observed ID-vs-OOD asymmetry (Guo et al. 2017; Ovadia et al. 2019), which we explicitly frame as an empirical prior rather than a theorem; the exploratory pilot result ($\Delta\text{ECE}_{\text{ID}} \approx 0$) only triggered the revision, did not choose the direction. The pilot results do not enter the main test report (protocol §7). All hypotheses in this paper are tagged *exploratory + robustness validation*, not confirmatory, to reflect the post-revision status.

6.7. *Limitations and future work*

Limitations: (i) the main endpoint is completed for InceptionTime and ResNet1D on all 6 directed transfer pairs with full 5-seed coverage (60 experiments total, fully balanced); the BiMamba architecture is included only as a toy experiment ($n = 60$, downsampled) deferred to the supplement (memory exhaustion on an RTX 5060, 8 GB) and does not enter the main-endpoint analysis; (ii) the predictability failure branch is triggered, downgrading the three-step chain to a two-step chain—the deployment criterion is an empirical lookup table, not a predictive criterion; (iii) the Shapley decomposition cross-fitting step remains to be run on real data; per the self-binding clause,

“mechanism” wording is limited to “mechanism decomposition (synthetic validation part)”; (iv) the 27-cell decomposition validation uses binned ECE, not the SmoothECE of the main endpoint; quantitative conclusions cannot be extrapolated across ECE definitions; (v) the safety criterion is rarely satisfied on the full L2 shift matrix (TS safety rate 0.0064), indicating that TS is a useful default but not a universally safe recalibrator; (vi) nine counter-examples out of 60 experiments (15.0%) are transparently disclosed, with a frequent post-hoc pattern (most counter-examples have low OOD accuracy, non-positive decay, and elevated OOD raw ECE) but no post-hoc exclusion; (vii) the operating CI is percentile rather than the pre-registered BCa, and $B = 10,000$ was used in 31/60 experiments and $B = 200$ in 21/60 (5 experiments lack log provenance); the BCa sensitivity analysis was re-run on 3 representative experiments only, with a full-grid BCa re-run committed for revision; (viii) the Youden deployment threshold is fitted in-sample (the protocol’s held-out dose levels were not used; the leave-shift-type-out recomputation shows the in-sample operating characteristic is optimistic); (ix) the L2 deployment/safety analysis covers 157 of 390 planned (pair, seed, shift) cells at 2 seeds on a single architecture, and the L2 noise levels are synthetic rather than the pre-registered PTB-XL Noise Stress Test Database (unavailable under PhysioNet licensing); (x) temperature was fitted once per experiment, so the reported CIs do not include temperature-fitting uncertainty (the pre-registered two-layer bootstrap is implemented but not enabled on real data); (xi) model selection used validation ECE rather than the pre-registered validation NLL early-stopping rule, which plausibly contributes to the positive ID boundary point estimates; (xii) the per-pair decomposi-

tion into covariate, prior-probability, and concept-shift components is not asserted—each directed pair combines all three shift types, and the L2 ladder isolates them only in eval-time; (xiii) the deployment lookup table is an audit-time tool whose inputs (raw ECE, ID–OOD accuracy gap, per-cell benefit labels) require labeled target data, and the label-free proxies tested in the predictability experiment failed their pre-registered branch, so no deployable label-free gate is claimed; (xiv) TS is argmax-invariant by construction and therefore cannot improve accuracy, AUROC, or AUPRC; per-direction discrimination metrics are not computed in the released artifacts (accuracy only), so the clinical viability precondition is screened by accuracy alone, and the measured OOD accuracy falls below the target majority-class baseline in 4 of 6 directions.

Future work: (1) cross-fitting validation of the Shapley decomposition on real data; (2) prospective multi-device validation of the deployment criterion; (3) extension to continuous shift spaces; (4) π -independent recovery estimators (BBSE/EM-class) to upgrade the MAPE_π from a design check to a real estimator test; (5) investigation of the direction-specific ResNet1D deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each) to determine whether it reflects a systematic architecture×direction interaction or random variation.

7. Conclusion

We pre-registered and validated a multi-seed calibration boundary study of temperature scaling in cross-corpus ECG transfer. The pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is supported in 51/60 seed experiments across

6 transfer pairs $\times$ 2 architectures (InceptionTime 27/30, ResNet1D 24/30, 85.0% aggregate), with nine counter-examples transparently disclosed and mechanistically explained. Architecture robustness is supported on both InceptionTime (27/30, 90.0%) and ResNet1D (24/30, 80.0%) with full 5-seed coverage on all 6 directions, enabling direct architecture comparison without seed-coverage confounding. The ID boundary criterion was *not* met (23/60 = 38.3% of cells have 95% CIs containing zero under the BCa operating CI, below the $\geq 80\%$ threshold; the percentile CIs gave 29/60 = 48.3%, same conclusion), so the pre-registered “ID nonzero boundary” branch was triggered: the OOD benefit is about $1.9\times$ the ID benefit on average and cannot be attributed exclusively to OOD-specific mismatch. The 27-cell Shapley decomposition validation passes the pre-registered threshold on identifiable cells. The method-selection boundary (TS robust; EM prior and Matrix scaling weakest, InceptionTime 15/30 each) is governed by a prior-likelihood mismatch mechanism. The predictability failure branch is triggered, downgrading the three-step chain to a two-step chain. All hypotheses are tagged exploratory + robustness validation, not confirmatory; the pre-registration protocol v2.1-A1 and five-item honesty declaration are archived for transparency. The findings suggest that TS is a reasonable default for cross-corpus ECG deployment, with two qualifications supported by the nine counter-examples: (1) most counter-examples co-occur with low OOD accuracy (≤ 0.50 in 6/9) and a large ID–OOD accuracy gap (≥ 0.18 in 7/9) — a heuristic *tendency*, not a sufficient gate (three counter-examples have OOD accuracy > 0.5 , so accuracy alone cannot rule out failure); (2) local validation is required (the TS safety rate 0.0064 under the operating point-estimate criterion on the

full L2 shift matrix shows that TS is not universally safe, and the Youden deployment criterion is fitted in-sample). Where these qualifications are not met, prior-correction methods require a confidence gate, and the deployment criterion should be combined with local validation. A third, precondition qualification applies throughout: recalibration presumes clinically acceptable discrimination, and the measured OOD accuracy is below the target majority-class baseline in 4 of the 6 directions—in those scenarios recalibration is moot until the discrimination problem is solved, because TS cannot alter accuracy by construction. All hypotheses are exploratory + robustness validation, not confirmatory.

Note on meta-analytic pooling. The meta-analytic pooled effect (fixed-effect inverse-variance weighting, see `meta_analytic_pooled.csv`) is sensitive to between-study heterogeneity in CI widths: the ResNet1D fixed-effect pooled estimate is slightly negative (-0.000573) because the narrow-CI studies dominate the inverse-variance weights. Under the DerSimonian–Laird random-effects model (also in `meta_analytic_pooled.csv`), all three strata are positive (InceptionTime $+0.0151$, ResNet1D $+0.0158$, CrossArchitecture $+0.0148$), and the simple unweighted means are likewise positive ($+0.0152$, $+0.0165$, $+0.0159$). The sign discrepancy between fixed-effect and random-effects pooling confirms that the fixed-effect pooled estimate is driven by CI-width heterogeneity ($\tau^2 \approx 3 \times 10^{-5}$, $Q \gg k - 1$) rather than by a genuinely negative central effect. We therefore report the **count-based sign proportion** ($51/60 = 85.0\%$ supported) as the primary, more robust test, and treat the meta-analytic pooled effect as a secondary, model-dependent summary.

Family size and multiple-comparison scope. The 60 seed experiments constitute the robustness-evidence family; the pre-registered confirmatory family is the shift-level grid of 6 pairs $\times$ 2 architectures $\times$ 13 shift levels = 156 comparisons (protocol §6). The 8-method design space (8 methods $\times$ 6 pairs $\times$ 5 seeds $\times$ 2 architectures = 480 cells) is reported in the method-selection boundary analysis (Sec. 5.3). Multiple-comparison corrections on the 60-seed robustness family leave the headline count unchanged (51/60 under BH-60 and Bonferroni), and the honest pair $\times$ architecture-stratum aggregation (12 strata) retains 9/12 directions after BH correction (InceptionTime 6/6).

CI width distribution. Across the 60 TS experiments, the 95% CI widths for $\Delta\text{ECE}_{\text{OOD}}$ range from 2.1×10^{-5} to 1.2×10^{-2} (median 1.2×10^{-3}). 52/60 TS cells fall below the repository’s narrow-CI diagnostic threshold (0.005), and six *supporting* cells have degenerate widths ($< 3 \times 10^{-4}$); excluding those six, the headline count becomes 45/60 (75.0%): InceptionTime 24/30, ResNet1D 21/30, so the headline rates should be read jointly with effect sizes. The narrow CIs partly reflect the moderate calibration sample size ($n_{\text{cal}} = 1027\text{--}2099$ records, 1027–1873 patients, per the per-experiment logs) and the low variance of the SmoothECE estimator; per-experiment bootstrap metadata (B, CI method) are not stored in the result files, which we flag as a provenance limitation. CI width does not affect the support/counter-example classification, which depends on the sign of the CI lower bound: a narrow CI around a positive point estimate yields a positive lower bound (support), while a narrow CI around a negative point estimate yields a negative lower bound (counter-example). The heterogeneity in CI widths across studies is

the reason the fixed-effect meta-analytic pooled estimate is model-sensitive (see note above), whereas the count-based sign proportion is not.

Honesty Declaration (5 items, adversarial audit product)

1. MAPE_π is a **design check** (resampling constant read-back), not an estimator capability test; independent π recovery (BBSE/EM) is a deferred item (protocol §13).
2. Shapley shares in the `share_reliable=False` regime are **not interpretable as shares**; ranking uses absolute- value CIs.
3. The threshold `gate(n)` is a heuristic scaling, **not a 3σ tolerance**; in the archived $n = 500$ seeded runs, 22/108 (20.4%) fail the gate (information mode), reported via exit code 2.
4. “Mechanism” wording is conditional on all three legs of §8 being fulfilled: legs 1–2 are fulfilled, leg 3 synthetic part (bootstrap CI + entanglement demo) is fulfilled; **cross-fitting on real data is pending** (registered in §13). The wording is limited to “mechanism decomposition (synthetic validation part)”.
5. The repository’s `scripts/preprocess_cinc2021.py`, `run_paper_benchmarks.py`, `resources/`, and `assets/` come from the third-party SafeECGMatch release (original README in `docs/vendor_safeecgmatch.md`), unrelated to this protocol’s experiment grid.

Data availability

The PTB-XL [3], Chapman-Shaoxing [4], and CPSC2018+2019 datasets are publicly available via PhysioNet/Kaggle under their respective licenses.

The pre-registered protocol v2.1-A1, its hash manifest (`docs/osf_archive_manifest.json`), the revision A1 document, and the full per-experiment result files (`checkpoints/transfer/**/transfer_results/*.csv`) are included with the submission as supplementary material; a public repository will accompany acceptance.

Declaration of competing interest

The authors declare no competing financial or personal interests.

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Table 1: Pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, $B = 10,000$, percentile operating CI). “Support” = 95% CI lower bound > 0 . Nine counter-examples are transparently disclosed. Full 5-seed coverage on both architectures for all 6 directions.

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
Chap $\rightarrow$ CPSC	Incep	42	+0.0075	+0.0139	+0.0135	✓
		43	+0.0012	+0.0101	+0.0099	✓
		44	+0.0104	+0.0186	+0.0181	✓
		45	+0.0019	−0.0057	−0.0058	✗
		46	+0.0044	+0.0130	+0.0126	✓
	ResNet	42	+0.0002	+0.0205	+0.0200	✓
		43	+0.0003	−0.0006	−0.0006	✗
		44	−0.0007	+0.0078	+0.0075	✓
		45	−0.0016	+0.0271	+0.0263	✓
		46	−0.0024	+0.0169	+0.0162	✓
Chap $\rightarrow$ PTB-XL	Incep	42	+0.0004	−0.0025	−0.0035	✗
		43	+0.0005	+0.0095	+0.0066	✓
		44	+0.0055	+0.0136	+0.0107	✓
		45	−0.0128	+0.0238	+0.0207	✓
		46	+0.0048	+0.0162	+0.0142	✓
	ResNet	42	−0.0031	+0.0309	+0.0285	✓
		43	+0.0128	+0.0291	+0.0285	✓
		44	+0.0059	+0.0172	+0.0168	✓
		45	−0.0010	−0.0036	−0.0036	✗
		46	−0.0042	+0.0149	+0.0139	✓
CPSC $\rightarrow$ Chap	Incep	42	+0.0029	+0.0065	+0.0062	✓
		43	−0.0012	+0.0209	+0.0200	✓
		44	+0.0164	+0.0182	+0.0160	✓
		45	+0.0091	+0.0245	+0.0241	✓
		46	+0.0014	+0.0024	+0.0023	✓
	ResNet	42	−0.0002	−0.0010	−0.0012	✗
		43	+0.0038	+0.0096	+0.0094	✓
		44	−0.0152	−0.0168	−0.0170	✗
		45	+0.0137	+0.0181	+0.0145	✓
		46	+0.0017	+0.0017	+0.0016	✓
		42	+0.0029	+0.0068	+0.0066	✓
		43	−0.0012	+0.0203	+0.0198	✓

Table 2: ID boundary condition ΔECE_{ID} for TS (12 pair $\times$ architecture strata, 5 seeds each). “CI $\cap$ 0” = number of seed cells whose 95% CI contains zero. The pre-registered boundary criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met: 23/60 cells (38.3%) contain zero (BCa operating CI) and 34/60 have CI lower bound > 0 , triggering the pre-registered “ID nonzero boundary” branch (protocol A1.4.2).

Transfer pair	Arch	Mean	Min CI lo	Max CI hi	CI $\cap$ 0/5
PTB-XL $\rightarrow$ CPSC	IT	+0.0130	−0.0121	+0.0266	1/5
PTB-XL $\rightarrow$ CPSC	RN	+0.0194	−0.0008	+0.0770	2/5
PTB-XL $\rightarrow$ Chap	IT	+0.0162	−0.0026	+0.0310	1/5
PTB-XL $\rightarrow$ Chap	RN	+0.0232	+0.0053	+0.0448	0/5
Chap $\rightarrow$ CPSC	IT	+0.0051	−0.0039	+0.0150	3/5
Chap $\rightarrow$ CPSC	RN	−0.0008	−0.0104	+0.0093	4/5
Chap $\rightarrow$ PTB-XL	IT	−0.0003	−0.0175	+0.0103	3/5
Chap $\rightarrow$ PTB-XL	RN	+0.0021	−0.0123	+0.0161	3/5
CPSC $\rightarrow$ Chap	IT	+0.0057	−0.0182	+0.0195	2/5
CPSC $\rightarrow$ Chap	RN	+0.0008	−0.0179	+0.0213	2/5
CPSC $\rightarrow$ PTB-XL	IT	+0.0046	−0.0181	+0.0192	2/5
CPSC $\rightarrow$ PTB-XL	RN	+0.0104	−0.0042	+0.0294	0/5
Overall		+0.0083			23/60 (38.3%)

Table 3: Method-selection boundary: 8 methods $\times$ 6 transfer pairs $\times$ 5 seeds (Inception-Time). Support rate = fraction of seeds with 95% CI lower bound > 0 . “Full” = number of directions with 5/5 full support. “Neg. dirs” = number of the 6 transfer directions whose 5-seed mean $\Delta\text{ECE}_{\text{OOD}}$ is negative. TS is the most robust parametric recalibrator; EM and Matrix scaling are the most fragile on InceptionTime (15/30 each); on ResNet1D, Matrix scaling (17/30) is the most fragile, EM reaches 19/30.

Method	Total	Full/6	$\Delta\text{ECE}_{\text{OOD}}$ mean	Neg. dirs
TS	27/30	3/6	+0.0152	0
Platt	24/30	3/6	+0.0169	0
Isotonic	19/30	2/6	+0.0100	2
Vector	21/30	1/6	+0.0158	0
Matrix	15/30	0/6	+0.0094	2
Dirichlet	20/30	1/6	+0.0139	1
EM prior	15/30	2/6	−0.0127	3
BBSE prior	22/30	3/6	+0.0424	1

Table 4: Architecture robustness for TS: InceptionTime vs. ResNet1D across 6 transfer pairs, full 5-seed coverage on both architectures. Architecture robustness is supported on both InceptionTime (27/30, 90.0%) and ResNet1D (24/30, 80.0%) with complete 5-seed coverage on all 6 directions. ResNet1D shows a direction-specific deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each), not a global architecture deficit.

Transfer pair	InceptionTime	ResNet1D	Total	Note
Chap→CPSC	4/5	4/5	8/10	2 counter-examples
Chap→PTB-XL	4/5	4/5	8/10	2 counter-examples
CPSC→Chap	5/5	3/5	8/10	2 counter-examples
CPSC→PTB-XL	5/5	3/5	8/10	2 counter-examples
PTB-XL→Chap	4/5	5/5	9/10	1 counter-example
PTB-XL→CPSC	5/5	5/5	10/10	zero counter-examples
Total	27/30	24/30	51/60	85.0%
Mean ΔECE	+0.0152	+0.0165	+0.0159	
Std ΔECE	0.0098	0.0180	0.0145	ResNet1D/Incep = 1.84
Mean OOD ECE	0.217 → 0.202 / 0.216 → 0.200	0.217 → 0.201	0.217 → 0.201	raw → TS

Table 5: Predictability LOO R^2 (multi-architecture merged). All methods trigger the failure branch (CI upper bound < 0.5). The three-step chain is downgraded to a two-step chain per the pre-registered failure branch.

Method	R^2	CI lower	CI upper	Verdict
TS	-0.096	-0.720	+0.104	Failure branch
Platt	-0.059	-0.584	+0.002	Failure branch
Isotonic	+0.011	-0.569	+0.101	Failure branch
Vector	-0.061	-0.522	+0.073	Failure branch
Matrix	-0.137	-0.516	-0.013	Failure branch
Dirichlet	-0.084	-0.394	+0.032	Failure branch
EM prior	+0.143	-0.250	+0.274	Failure branch
BBSE prior	+0.136	-0.354	+0.201	Failure branch

Table 6: Deployment criterion sensitivity / specificity at varying thresholds (raw ECE $>$ threshold $\rightarrow$ predict “deploy TS”). Optimal Youden $J = 0.124$ at threshold 0.127.

Threshold	Sensitivity	Specificity	TP	FP
0.094	0.976	0.067	444	723
0.107	0.956	0.138	435	668
0.127	0.923	0.201	420	619
0.178	0.807	0.288	367	552
0.277	0.499	0.508	227	381

Table 7: Trivial strategy comparison. Mean ΔECE ; negative = beneficial (ECE reduction). always-none and always-buy- n -labels span all 1230 (method, shift) cells; always-TS is evaluated on the 157 (pair, seed, shift) cells with complete results and lookup-table on 155 (two cells whose in-sample best method is unavailable are dropped). The lookup table captures most of the TS benefit without target labels.

Strategy	Mean ΔECE	Std	n
always-none	0.0000	0.0000	1230
always-TS	-0.0145	0.0099	157
always-buy- n -labels	-0.2905	0.1861	1230
lookup-table	-0.0160	0.0289	155